

Probability and Statistics (MT-2005)

Homework # 8

Normal Distribution & Central Limit Theorem

Questions

1. Cholesterol is a fatty substance that is an important part of the outer lining (membrane) of cells in the body of animals. Its normal range for an adult is 120–240 mg/dl. The Food and Nutrition Institute of the Pakistan found that the total cholesterol level for Pakistani adults has a mean of 159.2 mg/dl and 84.1% of adults have a cholesterol level below 200 mg/dl. Suppose that the total cholesterol level is normally distributed.
 - (a) Determine the standard deviation of this distribution.
 - (b) What are the quartiles (the 25% and 75% values) of this distribution?
 - (c) What is the value of the cholesterol level that exceeds 90% of the population?
 - (d) An adult is at moderate risk if cholesterol level is more than one but less than two standard deviations above the mean. What percentage of the population is at moderate risk according to this criterion?
 - (e) An adult is thought to be at high risk if his cholesterol level is more than two standard deviations above the mean. What percentage of the population is at high risk?
 - (f) An adult has low risk if cholesterol level is one standard deviation or more below the mean. What percentage of the population is at low risk?
2. The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed, suppose that the mean of the filling operation can be adjusted easily, but the standard deviation remains at 0.1 ounce.
 - (a) At what value should the mean be set so that 99.9% of all cans exceed 12 ounces?
 - (b) At what value should the mean be set so that 99.9% of all cans exceed 12 ounces if the standard deviation can be reduced to 0.05 fluid ounce?
3. An average scanned image occupies 0.6 megabytes of memory with a standard deviation of 0.4 megabytes. If you plan to publish 80 images on your web site, what is the probability that their total size is between 47 megabytes and 50 megabytes?
4. You take a sample of size $n = 49$ from a population and observe that the probability of the sample mean exceeding 55 is 0.05. The population standard deviation is known to be $\sigma = 14$. What is the population mean μ ?
5. Samples are taken from two populations:
 - Population A: $\mu_1 = 65$, $\sigma_1 = 10$, $n_1 = 50$
 - Population B: $\mu_2 = 70$, $\sigma_2 = 12$, $n_2 = 60$

What is the probability that the difference in sample means $(\bar{X}_1 - \bar{X}_2)$ is less than -3 ?

6. A population is heavily skewed, with:

- Mean $\mu = 120$
- Standard deviation $\sigma = 40$

A sample of size $n = 60$ is taken.

- (a) Can you apply the Central Limit Theorem to approximate the sampling distribution of $\bar{X}$?
- (b) Calculate the probability $P(\bar{X} > 130)$.